

Quang Truong

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Professional Summary

Doctoral student with 5+ years of experience in software engineering, machine learning, and artificial intelligence. Skilled in deploying scalable ML systems and deep learning solutions. Expertise in **Graph Machine Learning**, **Relational Deep Learning**, **Time-series Modeling**, and **Agentic AI**, with publications in **ICML**, **NeurIPS**, **AAAI**, and **JMLR**. Strong background in industry research, large-scale model development, and open-source contributions. Experienced across the ML lifecycle, including data processing, model design, training optimization, and distributed computing.

Work Experience

Snap Inc.

Bellevue, WA

Research Intern (Mentors: **Donald Loveland**, **Neil Shah**)

Jun. 2026 - Present

- Lead research projects in the user modeling and personalization domains (sub-areas include graph modeling, generative recommendation, personalization, and ML efficiency).
- Build scalable research prototypes and evaluate them in large-scale machine learning scenarios.

Nokia Corporation

Sunnyvale, CA

AI R&D Engineer Co-op (Mentors: **Liang Wu**, **Liangjie Hong**)

Jan. 2026 - May. 2026

- Designed, developed, and deployed advanced AI/ML models and algorithms to analyze and interpret complex data.
- Designed and implemented machine learning models to improve a wide range of applications, including search, forecasting, text mining, and more.
- Developed and implemented agentic-based systems for a wide range of applications, including anomaly detection, root-cause analysis, and more.

Education

Michigan State University

East Lansing, MI

Ph.D. in Computer Science

Aug. 2024 - May 2028

Dartmouth College

Hanover, NH

M.S. in Engineering Sciences

Sep. 2022 - Jun. 2024

Selected Publications

‡ indicates equal contribution.

Peer-reviewed Publications

- [1] Plain Transformers are Surprisingly Powerful Link Predictors
Quang Truong, Yu Song, Donald Loveland, Mingxuan Ju, Tong Zhao, Neil Shah, and Jiliang Tang. **2026**
ICML'26: The 43rd International Conference on Machine Learning.
- [2] A Pre-training Framework for Relational Data with Information-theoretic Principles
Quang Truong, Zhikai Chen, Mingxuan Ju, Tong Zhao, Neil Shah, and Jiliang Tang. **2025**
NeurIPS'25: The 39th Annual Conference on Neural Information Processing Systems.
- [3] TopoX: A Suite of Python Packages for Machine Learning on Topological Domains
Mustafa Hajji[‡], Mathilde Papillon[‡], Florian Frantzen[‡], ... **Quang Truong**, and ... Nina Miolane. **2024a**
The Journal of Machine Learning Research (Volume 25).
- [4] Weisfeiler and Lehman Go Paths: Learning Topological Features via Path Complexes
Quang Truong and Peter Chin. **2024b**
AAAI'24: The 38th Annual AAAI Conference on Artificial Intelligence.
- [5] Not All Data Matters: An Efficient Approach to Multi-Domain Learning in Vehicle Re-Identification
Quang Truong and Bo Mei. **2021**
ITSC'21: The 24th IEEE International Conference on Intelligent Transportation Systems.

Open-source Projects

TopoX

- Provides reliable and user-friendly building blocks for computing and machine learning on topological domains that extend graphs.
- Adopted open-source software engineering best practices with comprehensive testing, documentation, and CI/CD pipelines.
- Starred 500+ times across 3 different libraries on GitHub.
- Published at the Journal of Machine Learning Research (JMLR).

Research Experience

DSE Lab, Michigan State University

East Lansing, MI

Doctoral Research Assistant (Advisor: Jiliang Tang)

Aug. 2024 - Present

- Develop scalable transformers and characterize necessary conditions for the models to be strong Link Predictors (**Snap Research Collaboration**).
- Design and implement distributed training pipelines for large-scale graphs, optimizing performance across multiple GPUs and ensuring scalability for enterprise-scale data.
- Create automated testing frameworks and continuous integration pipelines to ensure code quality and system reliability across research projects.
- Develop specialized pre-training procedures for graph machine learning on relational databases, thereby enhancing predictive modeling for node-level classification, link prediction, and temporal forecasting tasks (**Snap Research Collaboration**).

LISP Lab, Dartmouth College

Hanover, NH

Graduate Research Assistant (Advisor: Peter Chin)

Sep. 2022 - Jun. 2024

- Introduced a novel approach to Topological Deep Learning (TDL) leveraging simple paths, significantly enhancing graph expressivity and surpassing the theoretical limitations of the 3-WL test.
- Developed Path Isomorphism Network, a state-of-the-art TDL model, which delivered outstanding performance on benchmark datasets: PROTEINS (78.8%), NCI1 (85.1%), NCI109 (84.0%), and IMDB-B (76.6%).
- Developed an anomaly detection pipeline for human behavior analysis using geospatially-tagged data, as part of the **IARPA-HAYSTAC** funded project.

Vision Lab, University of Illinois at Chicago

Remote

Research Intern (Advisor: Wei Tang)

May 2021 - Aug. 2021

- Designed a context-aware 3D object detection architecture that captures both object-wise and object-context relationships, improving spatial understanding.
- Boosted model performance through cross-modality alignment, improving the feature representation of objects.

ML Lab, Texas Christian University

Fort Worth, TX

Undergraduate Research Assistant (Advisor: Bo Mei)

May 2019 - May 2021

- Developed an advanced Vehicle Re-identification pipeline utilizing Generative Adversarial Networks (GANs) for adaptive domain learning, resulting in a 10% improvement in mean Average Precision (mAP) over the baseline.
- Engineered a novel filtering algorithm that reduces the training data requirement by 25% while maintaining model performance, significantly improving computational efficiency.

Proposal Writing

- On the Conditions for Transformers to be a Strong Link Predictor 2025
PIs: Hui Liu, Jiliang Tang
Role: Drafted part of the proposal
Result: ICML'26 publication and awarded research grant.

Skills

Programming	Python (Expert), C/C++ (Advanced), Java (Proficient), R (Proficient), MATLAB (Proficient)
Machine Learning	Graph Machine Learning, Relational Deep Learning, Knowledge Graphs, Recommender System, Large Language Models, Computer Vision, Data Mining
ML Frameworks	Pytorch, Pytorch Geometric, Pytorch Frame, NetworkX, TopoX, Pytorch Lightning, Hugging Face, Scikit-learn, Matplotlib, Numpy, Pandas
DevOps & Infrastructure	WandB, Docker, Singularity, Pytest, Amazon Web Services (AWS), Google Cloud Platform (GCP), Kubernetes, CI/CD, Monitoring, Distributed Systems
Software Engineering	Spring, Agile, JIRA, SCRUM, Git, Linux/Unix, MongoDB, DuckDB, MySQL, Unit Testing, Performance Optimization, Automation, Algorithm Design, Data Structures, System Design

Professional Services _____

Conference Reviewer

- Conference on Neural Information Processing Systems (NeurIPS) **2025, 2026**
- International Conference on Machine Learning (ICML) **2026**
- International Conference on Learning Representations (ICLR) **2024**
- Annual AAAI Conference on Artificial Intelligence (AAAI) **2025**
- Learning on Graphs Conference (LoG) **2025**

Journal Reviewer

- ACM Transactions on Knowledge Discovery from Data (TKDD)
- IEEE Transactions on Big Data (TBD)

Honors _____

- ICML’26 Gold Reviewer Award **2026**
- NeurIPS’25 Scholar Award **2025**
- VEF Fellow **2021**
- Vingroup Scholarship Recipient **2021**
- ACM-ICPC South Central Bronze Medal **2019**